

Executive Summary

This report evaluates a simple investment strategy that changes between U.S. Treasury EFT (TLT), Gold ETF (GLD) and S&P 500 ETF (SPY) according to the prevailing market volatility environment. The purpose is to determine whether adjusting investments to market conditions can achieve better growth with manageable risk than simply staying diversified or remaining invested in SPY.

The analysis used daily observations from November 18, 2004 to August 21, 2026 as the maximum common sample period for which all series have available data. Three broad market environments were identified i.e. low change environment, moderate change environment and a high change environment. Historically, SPY performed best in the low change environment, GLD in the moderate change environment and TLT in the high change environment.

When evaluated with a one-day implementation delay, the three-environment strategy produced an annualized return of 18.09%, annualized volatility of 15.52%, a return-to-risk ratio of 1.07 and a maximum drawdown of 24.01%. Performance exceeded that of the equal-weight portfolio in both return and risk-adjusted metrics, while drawdowns remained below those experienced by SPY.

These results are based on historical observations and do not guarantee future performance. However, they highlight the potential value of a systematic allocation framework that dynamically adapts to evolving market condition.

Key findings of the analysis

Finding	Result
Sample period	18 Nov. 2004 – 21 Aug. 2026
Usable daily observations	5,472
Preferred market environments	3
Preferred asset : Low-change environment	SPY
Preferred asset: Moderate-change environment	GLD
preferred asset : High-change environment	TLT
Annualized return of strategy	18.09%
Annual variability of strategy	15.52%
Largest historical decline	-24.01%

Step 1: Data Preparation and Exploration

The analysis uses daily market data from Yahoo Finance for TLT, GLD, SPY and the VIX. The common period begins when all four series have usable observations and ends at the latest common date. This produces consistent comparison across the three investment choices. The data checks showed a complete common sample with no missing values or duplicate dates after alignment.

Data quality item	Result
Common price-data period	18 Nov. 2004 – 21 Aug. 2026
Common price observations	5,473
Daily observations after calculating changes/returns	5,472
Missing values in common price sample	0
Duplicate dates	0

Figure 1 below shows that the three assets can move substantially from one trading day to another, particularly during periods of market stress. SPY shows the largest day-to-day swings, while TLT tends to be less volatile over many periods. GLD occupies an intermediate position, although it can also experience large movements during stressed markets.

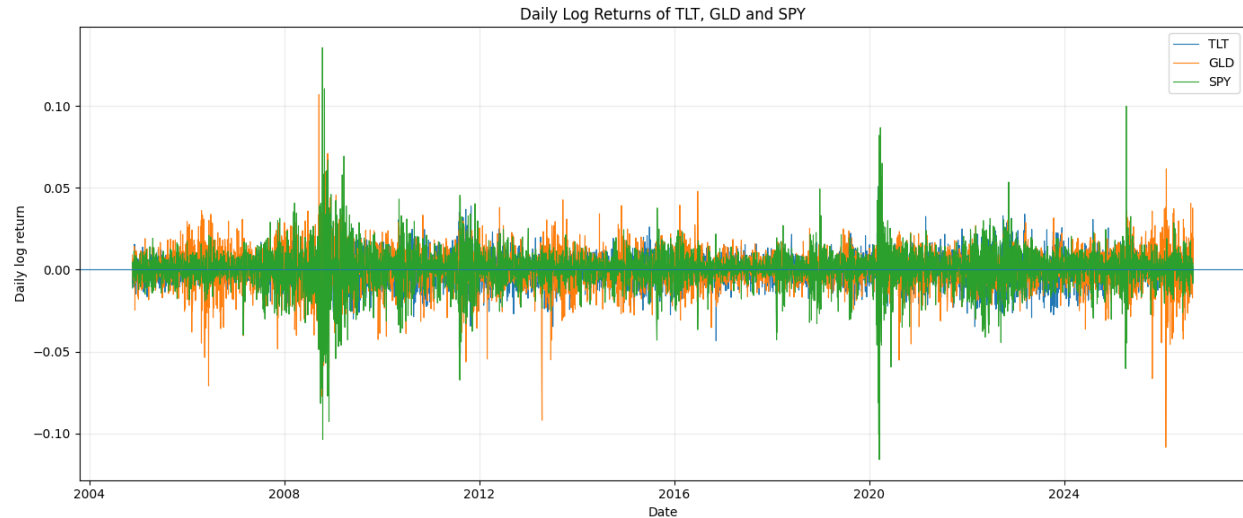


Figure 1: Daily movements of TLT, GLD and SPY over the sample period

Figure 2 shows that most of daily VIX changes are relatively small but there are occasional sharp increases. These large movements are concentrated around periods of market stress and provide the basis for distinguishing calm conditions from unsettled ones.

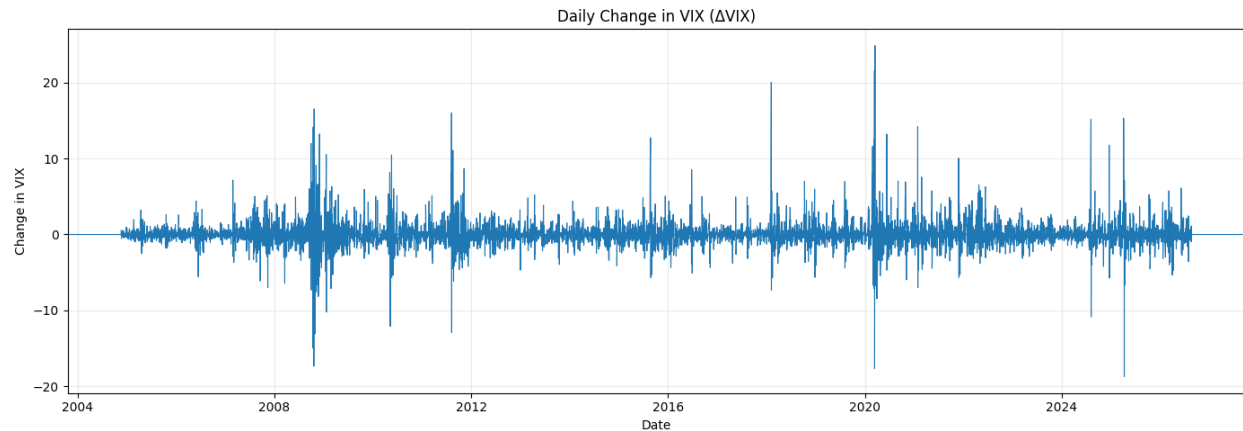


Figure 2: Daily change in the VIX (used as an indicator of the changing market volatility)

Step 2: Market Volatility Environments (VIX Regimes)

The VIX movements were grouped into three recurring environments. The environments are ordered from the least unsettled (low-change) to the most unsettled (high-change) conditions. The high-change environment occurs less frequently than the other two, which is consistent with severe volatility episodes being relatively uncommon.

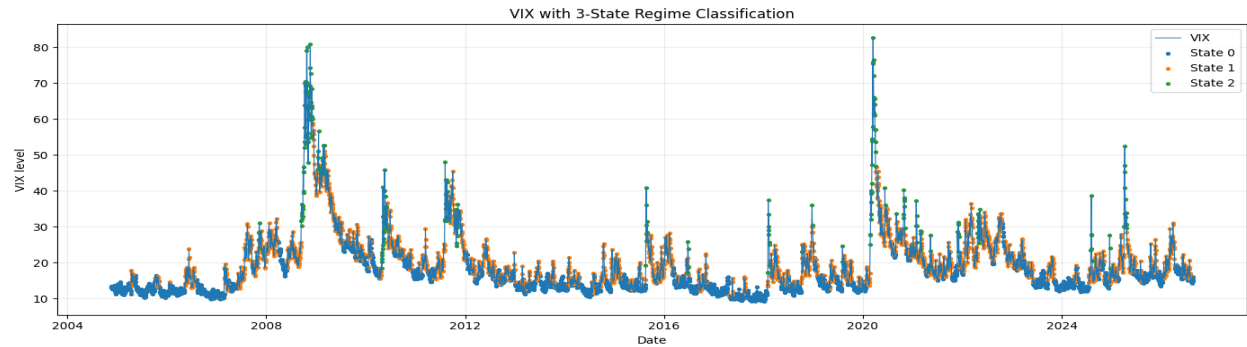


Figure 3: VIX level with observations classified into three volatility environments

The high-change environment is the most distinct with a positive average VIX movement and its day-to-day variation is much larger than in other environments. This is the environment in which the market experiences the strongest volatility pressure.

Environment	Typical VIX change	Variability of VIX change	Trading days	Investment interpretation
Low change	-0.0621	0.6203	2,905	Relatively calm conditions
Moderate change	-0.0110	1.7630	2,308	More mixed conditions
High change	+0.6554	6.0280	259	Strong volatility pressure

To the point made earlier, high-change environments are relatively uncommon as illustrated in the chart below;

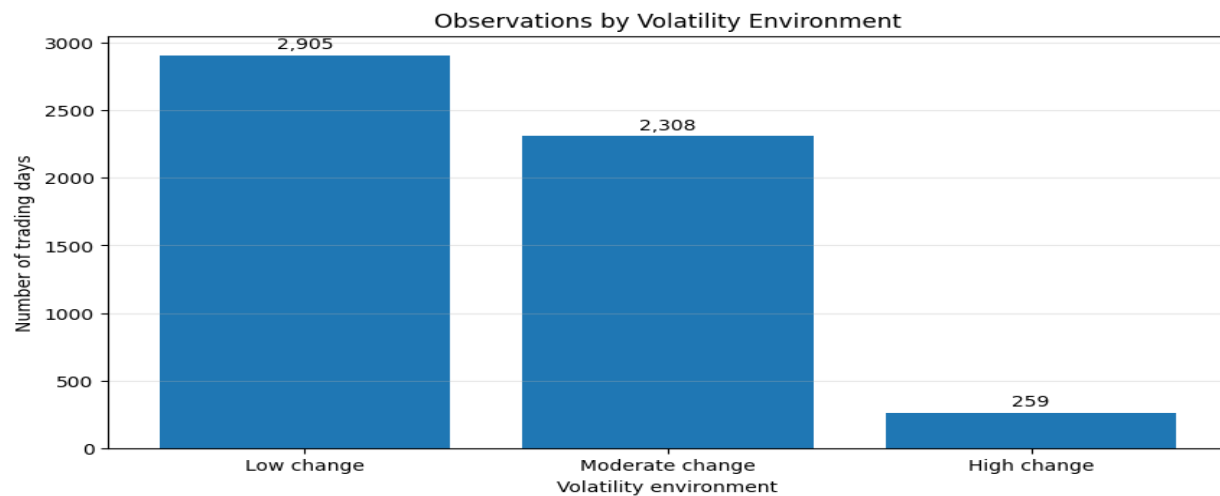


Figure 4: Number of trading days represented by each volatility environment

Step 3: State Selection and Interpretation

The three-state HMM is preferred because it has a higher log-likelihood and lower AIC and BIC than the two two-state HMM. All the three yardsticks support the use of the three-market environment for the remainder of the analysis.

Model	States	Parameters	Log-Likelihood	AIC	BIC
2-State HMM	2	7	-9,318.834	18,651.668	18,697.705
3-State HMM	3	14	-8,971.498	17,970.995	18,063.677

With the three-state HMM chosen as the preferred model, the average daily performance of TLT, GLD and SPY was compared. The purpose was to identify the asset which historically offered the strongest average outcome when each type of market condition prevailed.

State	TLT Mean	TLT Std.	GLD Mean	GLD Std.	SPY Mean	SPY Std.
0	-0.017%	0.743%	0.047%	0.953%	0.134%	0.564%
1	0.031%	0.962%	0.034%	1.235%	-0.004%	1.237%
2	0.152%	1.799%	0.036%	2.053%	-0.590%	3.498%

The results also showed a clear change in relative performance i.e. in calm conditions, the SPY has the strongest average return. In the moderate environment, the GLD has the strongest average return, while the SPY is negative. In the high-change environment, the TLT has the strongest average return and the SPY records a substantially negative average return.

This framework provides the economic rationale for rotating among the three assets rather than maintaining the same exposure throughout the entire period.

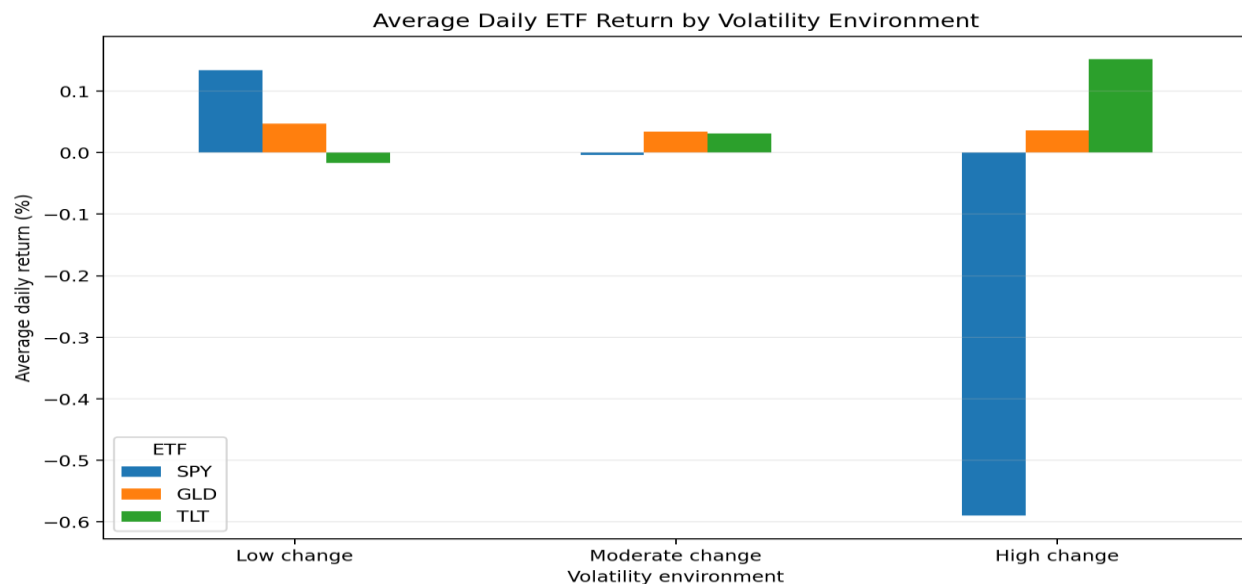


Figure 5: Average daily EFT returns within each volatility environment

Step 4: Designing the Rotation Strategy

Once the current market environment is identified, the investment rule is that the portfolio is assigned to entirely to the asset that historically performed best (highest historical mean return) in that environment. A one-day trading delay is then applied before the position is implemented.

Volatility environment	Portfolio allocation	Reason
Low change	100% SPY	SPY had the highest historical average return.
Moderate change	100% GLD	GLD had the highest historical average return.
High change	100% TLT	TLT had the highest historical average return.

The one-day delay ensures that the strategy only changes the portfolio after the market environment has been identified, rather than using information from the same day to explain or react to that day's return

Step 5: Backtesting and Evaluation

The strategy is compared with two alternatives, the first invests one-third in each of the assets rebalanced monthly and the second is to remain invested in SPY throughout the period. The alternatives were compared on total growth, annual growth, variability of returns, returns relative to variability and the largest decline from a previous peak.

Portfolio	Cumulative Return	Annualized Return	Annual volatility	Sharpe Ratio	Largest decline
Regime-based strategy	13.9898%	0.1328	0.1576	0.7910	-0.3522
Equal-weight portfolio	4.5632%	0.0822	0.9813	0.8054	-0.2346
SPY buy-and-hold	8.6021%	0.1098	0.1889	0.5515	-0.5519

The regime-based strategy has the highest annual growth of the three portfolios in the historical sample. It also had a better return relative to variability than either benchmark. Its annual variability is higher than that of the equal-weight portfolio but lower than that of the SPY buy-and-hold.

The regime-based strategy's largest decline was 35.22% compared with 23.46% for equal-weight portfolio and 55.19% for SPY buy-and-hold. This shows that the strategy historically avoided some of the deeper losses experienced by remaining fully invested in SPY.

The cumulative performance chart shows the growth of an \$1 investment over the sample period. The regime-based strategy finishes substantially above both comparison portfolios, consistent with its higher annual growth over the historical period.

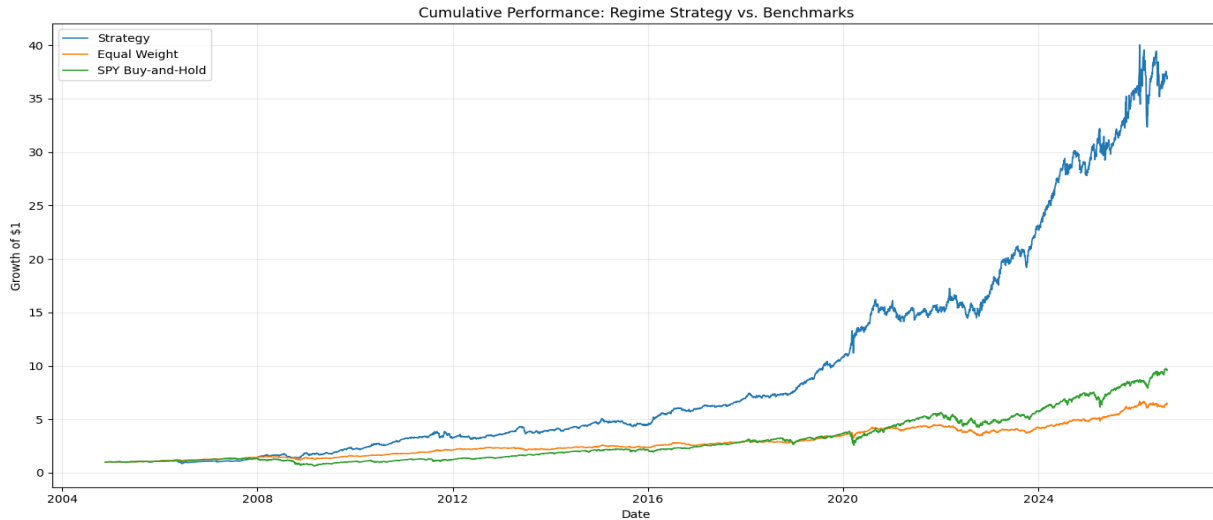


Figure 6: Cumulative performance of the regime-based strategy and the two benchmarks

The drawdown chart shows the decline experienced by each portfolio from previous highs. The deepest decline are more pronounced for SPY buy-and-hold, while the regime-based strategy generally experiences smaller declines.

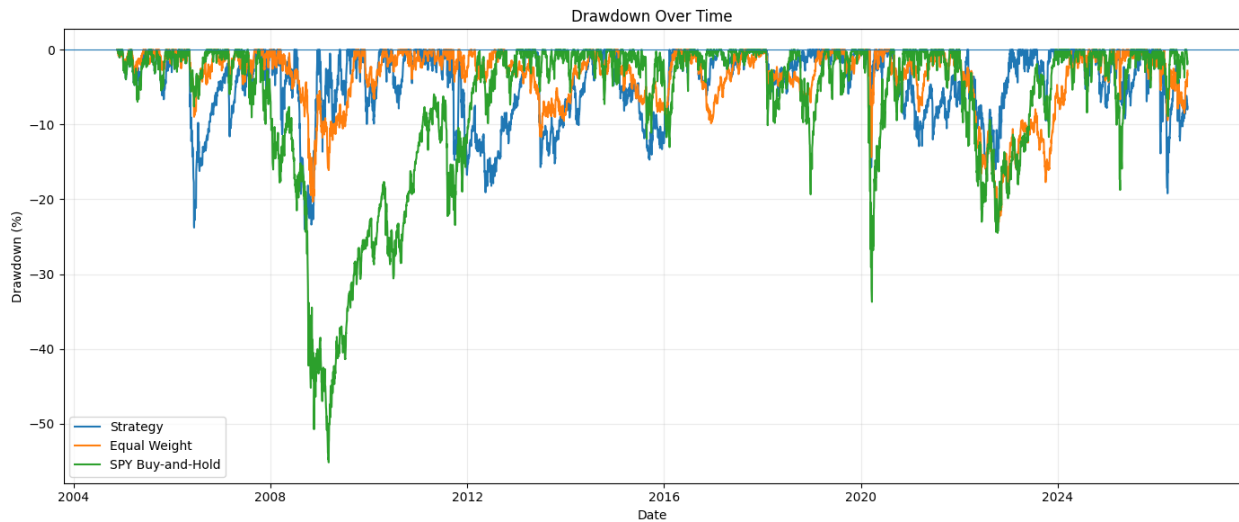


Figure 7: Historical drawdown of the regime-based strategy and the two benchmarks

Sensitivity/Stability of Investment Decision

A sensitivity analysis was done to check whether using two rather than three market environments materially changes the investment results.

Number of environments	Cumulative Returns	Annualized Return	Annual variability	Return / variability	Largest decline
2 environments	5214.69	20.08%	15.15%	1.21	-22.15%
3 environments	1398.98	13.28%	15.76%	0.79	-35.22%

The two-environment approach produced stronger historical results across all four measures. However, the three-environment approach provides a clearer distinction between moderate and high-change condition, allowing the portfolio to allocate GLD in moderate-change conditions and TLT in high-change

conditions. The three-environment approach was therefore retained as the preferred strategy for this analysis.

Other Factors Outside the Analysis that could affect the Investment

The allocation rule is based on historical performance under different market conditions. In practice, the underlying investments can also be affected by broader economic factors. These are not additional variables to the backtest rather they are considerations for interpreting and monitoring the investment choices.

These factors explain why no single investment is consistently the strongest choice across every environment. The allocation framework therefore responds to changing conditions rather than assuming that one asset will always be the best choice.

Investment	Important factors to monitor	Implication for the allocation decision
SPY	U.S. economic growth, company earnings, equity-market sentiment and market stress	More relevant when market conditions are relatively calm and equity performance is strong.
GLD	Gold prices, interest-rate conditions, inflation expectations and geopolitical developments	Can become more attractive when conditions are mixed and investors seek an alternative to equities.
TLT	U.S. interest rates, inflation expectations, monetary policy and demand for long-term government bonds	Can become more attractive when market stress is high and demand for government bonds increases.

Overall Conclusion

The historical evidence supports a disciplined approach in which the portfolio changes according to the prevailing market environment. The clearest investment pattern is SPY in low-change conditions, GLD in moderate conditions and TLT during high-change conditions.

For the project, the three-environment allocation was recommended as the primary strategy because it produced the strongest annual growth among the three portfolios, a better relative return to variability than both benchmarks and a substantially lower largest decline than SPY buy-and-hold.

Nevertheless, the results should be treated as historical evidence rather than a promise of future performance.

References

hmmlearn Developers. *hmmlearn Documentation*. Documentation for Gaussian hidden Markov models, <https://hmmlearn.readthedocs.io/>.

WorldQuant University. *MScFE 622 Stochastic Modeling – Group Work Project #2*. Assignment brief and grading rubric.

WorldQuant University. *MScFE 622 Stochastic Modeling: Lesson Notes*. WorldQuant University.

Yahoo Finance. *Yahoo Finance*. Data source for daily market prices and VIX observations, <https://finance.yahoo.com/>.